

# EDA Notebook

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## 0. Setup Environment

```
# DO NOT MODIFY THE CODE IN THIS CELL
!pip install -q utstd

from utstd.folders import *
from utstd.ipyrenders import *

at = AtFolder(
    course_code=36106,
    assignment="AT3",
)
at.run()

import warnings
warnings.simplefilter(action='ignore')
```

---

```
12.9/12.9 MB 48.2 MB/s eta
0:00:00
```

---

```
4.9/4.9 MB 37.3 MB/s eta
0:00:00
```

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

umap-learn 0.5.12 requires scikit-learn>=1.6, but you have scikit-learn 1.5.2 which is incompatible.

hdbscan 0.8.42 requires scikit-learn>=1.6, but you have scikit-learn 1.5.2 which is incompatible.

Mounted at /content/gdrive

You can now save your data files in:  
/content/gdrive/MyDrive/36106/assignment/AT3/data

---

## Student Information

```
# <Student to fill this section>
group_name = "Group 20"
student_name = "Ratnadeep Patra"
student_id = "26294153"

# Do not modify this code
print_tile(size="h1", key='group_name', value=group_name)
```

```
<IPython.core.display.HTML object>

# Do not modify this code
print_tile(size="h1", key='student_name', value=student_name)

<IPython.core.display.HTML object>

# Do not modify this code
print_tile(size="h1", key='student_id', value=student_id)

<IPython.core.display.HTML object>
```

---

## 0. Python Packages

### 0.a Install Additional Packages

If you are using additional packages, you need to install them here using the command: `! pip install <package_name>`

```
# <Student to fill this section>
```

### 0.b Import Packages

```
# <Student to fill this section>
import pandas as pd
import altair as alt

import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import os
```

---

## B. Data Understanding

```
# Do not modify this code
try:
    df = pd.read_csv(at.folder_path / "store.csv")
except Exception as e:
    print(e)
```

### B.1 Explore Dataset

```
df.describe(include='all').T

{"summary": "{\n  \"name\": \"df\",\n  \"rows\": 3,\n  \"fields\": [\n    {\n      \"column\": \"count\",\n      \"properties\": {\n
```

```

{"dtype\\": \\\"date\\\",\\n          \\\"min\\\": \\\"701\\\",\\n          \\\"max\\\": \\\"701\\\",\\n          \\\"num_unique_values\\\": 1,\\n          \\\"samples\\\": [\\n          \\\"701\\\"\\n          ],\\n          \\\"semantic_type\\\": \\\"\\\",\\n          \\\"description\\\": \\\"\\\"\\n          }\\n          },\\n          {\\n          \\\"column\\\": \\\"unique\\\",\\n          \\\"properties\\\": {\\n          \\\"dtype\\\": \\\"date\\\",\\n          \\\"min\\\": 13,\\n          \\\"max\\\": 701,\\n          \\\"num_unique_values\\\": 3,\\n          \\\"samples\\\": [\\n          701\\n          ],\\n          \\\"semantic_type\\\": \\\"\\\",\\n          \\\"description\\\": \\\"\\\"\\n          }\\n          },\\n          {\\n          \\\"column\\\": \\\"top\\\",\\n          \\\"properties\\\": {\\n          \\\"dtype\\\": \\\"string\\\",\\n          \\\"num_unique_values\\\": 3,\\n          \\\"samples\\\": [\\n          \\\"fb3f2681-31a3-4309-8654-088e5075dca7\\\"\\n          ],\\n          \\\"semantic_type\\\": \\\"\\\",\\n          \\\"description\\\": \\\"\\\"\\n          }\\n          },\\n          {\\n          \\\"column\\\": \\\"freq\\\",\\n          \\\"properties\\\": {\\n          \\\"dtype\\\": \\\"date\\\",\\n          \\\"min\\\": \\\"1\\\",\\n          \\\"max\\\": \\\"80\\\",\\n          \\\"num_unique_values\\\": 3,\\n          \\\"samples\\\": [\\n          \\\"1\\\"\\n          ],\\n          \\\"semantic_type\\\": \\\"\\\",\\n          \\\"description\\\": \\\"\\\"\\n          }\\n          }\\n          ]\\n          }\", \"type\": \"dataframe\"}

```

```
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 701 entries, 0 to 700
Data columns (total 3 columns):
#   Column          Non-Null Count  Dtype
---  -
0   store_id        701 non-null    object
1   sales_person_id 701 non-null    object
2   name            701 non-null    object
dtypes: object(3)
memory usage: 16.6+ KB

```

```
df.sample(5)
```

```

{"summary": "{\\n  \\\"name\\\": \\\"df\\\",\\n  \\\"rows\\\": 5,\\n  \\\"fields\\\": [\\n  {\\n    \\\"column\\\": \\\"store_id\\\",\\n    \\\"properties\\\": {\\n    \\\"dtype\\\": \\\"string\\\",\\n    \\\"num_unique_values\\\": 5,\\n    \\\"samples\\\": [\\n    \\\"9a0ba015-80b1-4df5-a54a-58c3676f6149\\\",\\n    \\\"57333052-bffd-437b-b0e2-f27052e0e58f\\\",\\n    \\\"552c7546-dc57-4509-b2d7-472f901fbf71\\\"\\n    ],\\n    \\\"semantic_type\\\": \\\"\\\",\\n    \\\"description\\\": \\\"\\\"\\n    }\\n    },\\n    {\\n    \\\"column\\\": \\\"sales_person_id\\\",\\n    \\\"properties\\\": {\\n    \\\"dtype\\\": \\\"string\\\",\\n    \\\"num_unique_values\\\": 3,\\n    \\\"samples\\\": [\\n    \\\"48505c49-f9c7-4476-a670-158f89c331a7\\\",\\n    \\\"27d62980-8d18-4170-95a5-7f7bb9cada5a\\\",\\n    \\\"28141e19-49e8-4090-9196-6e8f43b10de8\\\"\\n    ],\\n    \\\"semantic_type\\\": \\\"\\\",\\n    \\\"description\\\": \\\"\\\"\\n    }\\n    },\\n    {\\n    \\\"column\\\": \\\"name\\\",\\n    \\\"properties\\\": {\\n    \\\"dtype\\\": \\\"string\\\",\\n    \\\"num_unique_values\\\": 5,\\n    \\\"samples\\\": [\\n    \\\"Tire Exchange\\\",\\n    \\\"Some Discount Store\\\",\\n

```



```
9737ee67-ea97-4ab7-9e6c-70f8878c6ab2    40
f077eda9-c3db-4fde-81d2-e0f7ada08fcc    40
0445ab08-f703-4d2c-8ab3-fb7948bc1a48    40
16eb6889-25f7-475a-9eff-92024a00fde9    39
e03321bf-f1d3-4324-9184-008d6fb535da    38
48505c49-f9c7-4476-a670-158f89c331a7    38
Name: count, dtype: int64
```

```
(
    df['sales_person_id'].nunique(),
    df['sales_person_id'].value_counts().mean(),
    df['sales_person_id'].value_counts().max(),
    df['sales_person_id'].value_counts().min(),
)
```

```
(13, np.float64(53.92307692307692), 80, 38)
```

```
# <Student to fill this section>
```

```
feature_1_insights = """
```

```
The 'sales_person_id' feature indicates a highly concentrated
distribution, with only 13 unique salespersons managing 701 stores. On
average, each
```

```
salesperson is responsible for approximately 53.92 stores. The
salesperson with the highest workload is assigned 80 stores, while the
minimum is 38. This
```

```
suggests an organizational structure where a few sales personnel
manage a significant number of retail locations. Further insights into
the
```

```
'sales_person_id' can be gained from the 'sales_order_header.csv'
file, which likely contains transaction-level data related to these
salespersons.
"""
```

```
# Do not modify this code
```

```
print_tile(size="h3", key='feature_1_insights',
value=feature_1_insights)
```

```
<IPython.core.display.HTML object>
```